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The Rise Of The Body Bots

EXOSKELETONS ARE STRUTTING out of the lab—and they are carrying their creators with them. Erico Guizzo and Harry Goldstein of the IEEE Spectrum give us an insight into this future possibility

Science-fiction fans have long become accustomed to the idea of steely commandos clad in robotic exoskeletons taking on huge, vicious, extraterrestrial beasts, shadowy evil cyborgs, or even each other. Supersoldiers encased in sleek, self-powered armor figure memorably in such works as Robert A. Heinlein's novel *Starship Troopers*, Joe W. Haldeman's *The Forever War*, and many other books and movies. In *A Good Old-Fashioned Future*, for example, Bruce Sterling writes of a soldier dying after crashing in his "power-armor, a leaping, brick-busting, lightning-spewing exoskeleton."

Today, in Japan and the United States, engineers are finally putting some practical exoskeletons through their paces outside of laboratories. But don't look for these remarkable new systems to bust bricks or spew lightning. The very first commercially available exoskeleton, scheduled to hit the market in Japan next month, is designed to help elderly and disabled people walk, climb stairs, and carry things around. Built by Cyberdyne Inc., in Tsukuba, Japan, this exoskeleton, called HAL-5, will cost about 1.5 million yen (around US \$13 800).

Meanwhile, in the United States, the most advanced exoskeleton projects are at the University of California, Berkeley, and at Sarcos Research Corp., in Salt Lake City. Both are funded under a \$50 million, five-year program begun by the Defense Advanced Research Projects Agency, or DARPA, in 2001. During the past several months, each group has been working on a second-generation exoskeleton that is a huge improvement over its predecessor. Little information about the new models had been officially released by press time, but IEEE Spectrum has learned that the Berkeley unit was successfully tested in a park near the campus this past summer and the latest Sarcos model was demonstrated to a panel of military observers at Fort Belvoir, VA., last April.

<http://webench.national.com>

HAL-5, in Japan, and the systems by Berkeley and Sarcos, in the United States, appear to be the first of a platoon of considerably more capable exoskeletons aimed at real-world uses that may soon, quite literally, be walking near you.

Most of these systems are designed to help physically weak or injured people gain more mobility or perform rehabilitation exercises. But researchers are quick to mention other commercial possibilities for their creations: rescue and emergency personnel could use them to reach over debris-strewn or rugged terrain that no wheeled vehicle could negotiate; firefighters could carry heavy gear into burning buildings and injured people out of them; and furniture movers, construction workers, and warehouse attendants could lift and carry heavier objects safely.

For the entire article go to <http://www.spectrum.ieee.org/oct05/1901>



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Battling Bugs: A Digital Quagmire

HOW EFFECTIVE ARE our computer bug battling techniques? Why is it that we slip up in the same areas all the while? This article by Simson Garfinkel tries to find the reasons

In 1976, computer pioneer Edsger W. Dijkstra made an observation that would prove uncanny: "Program testing can be quite effective

for showing the presence of bugs," he wrote in an essay, "but is hopelessly inadequate for showing their absence."

Thirty tears later, Dijkstra's words have the ring of prophecy. Companies like Microsoft and Oracle, along with open-source projects like Mozilla and Linux, have all instituted rigorous and

extensive testing programs, but bugs just keep slipping through. Last month, Microsoft's monthly drop of bug patches included fixes for 14 security holes that escaped prerelease testing, four of them rated "critical."

On Tuesday, the company fixed three more Windows bugs, and all three were the same basic genus of bug—the "buffer overflow"—that helped spread the first Internet worm in 1988. It seems programmers and software architects manage to make the same mistakes generation after generation. Even back in 1988, many of the bugs that haunt us today were already old hat.

"We solved buffer overflows and the Y2K problem with Multics in 1975," says Peter Neumann, a senior scientist at SRI International who has been researching bugs and their impact on society for more than two decades. But while Multics—the first secure multi-user operating system—addressed some thorny problems, bug history keeps repeating itself.

The reasons for that are both simple and complex, experts say, having to do with the programming languages themselves or with programmer psychology and the environment in which software is developed. To understand why bugs occur, it helps to start by looking at the general classes of faulty code.

Bugs can be broadly divided into two categories. Typographical bugs and errors in reasoning are one type. Then, there are the deep, conceptual bugs that make a program malfunction even though all the code is more or less correct.

Memory misdeeds

Buffer overflows and race conditions are examples of the first kind of bug. A particularly tenacious beast, the potential for a buffer overflow is created when a programmer allocates a certain amount of memory to hold a piece of information—for example, nine characters to hold a Social Security number. But then the program tries to store more data in that space when it actually runs. The rest of the data overflows the pre-allocated buffer and overwrites something else in the computer's memory—frequently with disastrous results.

The 1990s saw buffer overflows reach near-epidemic proportions in programs written in the C and C++ programming languages, because these languages require coders to manually manage the memory used by their programs. Like driving a performance car, control of the memory might let a skilled programmer eke a bit more out of the computer, or accomplish neat tricks and stunts. But the danger of a stall or a crash is ever present.

For the entire article go to

http://wired.com/news/technology/bugs/0,2924,69369,00.html?tw=wn_16techhead

